1. What is a function in Python and why are functions useful in programming?

A function is a reusable block of code that performs a specific task. Functions are used to break down the complex problem into smaller, manageable, read,sub-problem making code easier, debug and maintain. It allows you to define code once and execute multiple times, avoiding redundant code.

2. How do you define a function in Python? Write the syntax with a simple example.

Function is defined by the **def** keyword. It take input by passing parameters.

Syntax:

```
def student_name():
   print('I am Tom')
   return student_name
student_name()
```

I am Tom

3. Explain the difference between parameters and arguments in the context of Python functions.

Parameters are the variable names that are defined in the parentheses in a function's definition. Arguments are the values passed during the function call.
def add(x,y) # x & y are parameters
Result add(3,5) #3 & 5 are arguments

4. What does the return statement do in a function? Give an example.

Return statement immediately exits the function and sends back the value to where function was called.

```
def result():
   result = 5+3
   return result
print(result())
```

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5. Can a Python function return multiple values? If yes, how?
Function can return multiple values by comma separated values. These values are usually returned as **tuple.**

1. Write a function that accepts a list of numbers and returns the sum of all the numbers.

```
def sum_list(numbers):
   sum = 0
   for x in numbers:
       sum += x
       return sum
print('Total value of list of num :',sum((25,36,86,64,21)))
```

```
Total value of list of num : 232
```

2. Write a function takes a number as input and returns True if the number is Even otherwise return False

```
def even(num):
   if num % 2 == 0:
       return True
   else:
       return False
print(even(10265))
print(even(225))
print(even(1264))
```

```
D:\SkillIT\Python\.venv\Scripts\python.
False
False
True
```

3. Write a function takes a string as input and returns the same string with all characters in Uppercase

```
name = 'sravanthi'
def student(name):
   return name.upper()
print(student(name))
```

```
D:\SkillIT\Python\.venv\Scr
SRAVANTHI
```

4. Write a function takes a string as input and returns the same string with all characters in lowercase

```
name = 'NETFLIX'
def student(name):
   return name.lower()
print(student(name))
```

```
:
D:\SkillIT\Python\.venv\Scripts\py
netflix
```

5. Write a function that takes 2 strings as input and returns one string which is concatenated of those 2 strings

```
n1 = 'ABC'
n2 = 'DEF'
def func1():
   string = n1 + n2
   return string
print(func1())
```

```
D:\SkillIT\Python
ABCDEF
```

6. Write a function that takes 2 strings as input and returns one string which is concatenated of those 2 strings with * in between them

```
n1 = 'ABC'
n2 = 'DEF'
def func1():
   string = n1 +'*' + n2
   return string
print(func1())
```

```
:
D:\SkillIT\Python\.venv\Scripts\
ABC*DEF
```

7. Write a function that takes 2 strings as input and returns one string which is concatenated of those 2 strings in lowercase and with * in between them

```
n1 = 'ABC'
n2 = 'DEF'
def func1():
   string = n1 +'*' + n2
   return string.lower()
print(func1())
```

```
abc*def
```

8. Write a function that takes list of string values and returns a string concatenated with all the elements in it separated by comma ‘,’

```
lst = ['A','B','C','D','E','F']
def func(string_list):
   result = ''
   for i in range(len(string_list)):
       result += string_list[i]
       if i < len(string_list) - 1:
           result += ','
   return result

print(func(lst))
```

```
D:\SkillIT\Python\.venv\Script
A,B,C,D,E,F

Process finished with exit cod
```